Algebra & Functions — Timed Simulation 1

IBEW / ETA math section format

33	46	~84 sec	No
questions	minutes	per question	calculator

About this test

- All 33 questions from your photos of Timed Simulation 1, retyped clean and printable, in the original order. Wording, numbers and answer choices are kept as they appeared on screen.
- Graphs are redrawn on a clean grid to match what was on screen.
- Multiple choice, A–D. No calculator. No penalty for guessing — answer everything.

How to take it

- Set a timer for 46 minutes and don't stop it.
- Skip anything that takes more than about 90 seconds and come back to it.
- The answer key with a short explanation for every question is at the back.

What's on it

Simplifying, expanding & factoring	Questions 3, 14, 15, 16, 17
Number series	Questions 8, 9, 10, 11, 12, 13
Plugging in & solving	Questions 1, 2, 25, 26, 27, 28, 31
Isolating a variable	Questions 32, 33
Variable relationships (which statement is true)	Questions 19, 20, 21, 22, 23, 24
Same relationship: tables, graphs, words	Questions 4, 5, 6, 7
Graphing & systems	Questions 18, 29, 30

Question 1

Consider the following formula:

$$C(B-3)=50-(B+A(B+1))$$

If $A=2$ and $C=10$, what is the value of B?

A) 2

B) 6

C) 11

D) 13

WORKSPACE

Question 2

Consider the following formula:

$(3A - (2B + C)) - 2 = BC + A$

If $B = 5$ and $C = 6$, what is the value of A?

A) 6

B) 12

C) 16

D) 24

WORKSPACE

Question 3

Simplify the following expression to its simplest form.

Consider $B = 5$ and $C = -1$:

$$\frac{A^3 + A}{A} + (C + B)(B - C) - A^2$$

A) 25

B) $A^3 - A^2 + 24$

C) $2A^2 - 23$

D) $A + 24$

WORKSPACE

Question 4

Consider the following function:

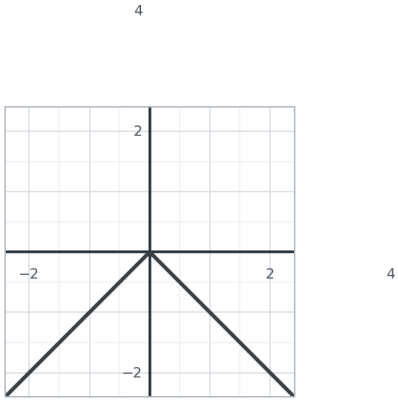
$y = -x + 1$

Which of the following choices represents the same relationship as the given function?

A)

x	y
−2	1
−1	0
0	−1
1	−2
2	−3

B)



C)

x	y
1	0
10	−9
100	−99
1000	−999

D) Y is equivalent to the value of X minus the value of X plus one.

WORKSPACE

Question 5

Consider the following information:

Y is equivalent to ten times the value of X divided by five.

Which of the following choices represents the same relationship as the aforementioned information?

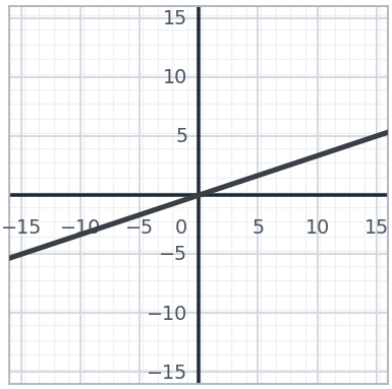
A)

x	y
0	0
1	2
2	4
3	6
4	8

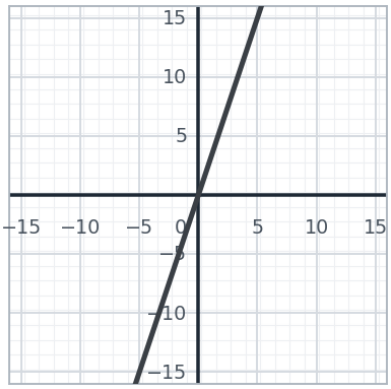
B)

x	y
−2	−25
−1	−15
0	−5
1	5
2	15

C)



D)



WORKSPACE

Question 6

Consider the following table:

x	y
0	−3
2	−1
3.5	0.5
6	3
8	5

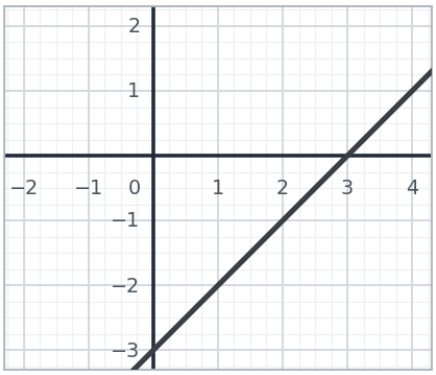
Which of the following choices represents the same relationship as the given table?

A) Y is equivalent to the difference between the value of X and a constant C, where C equals −3.

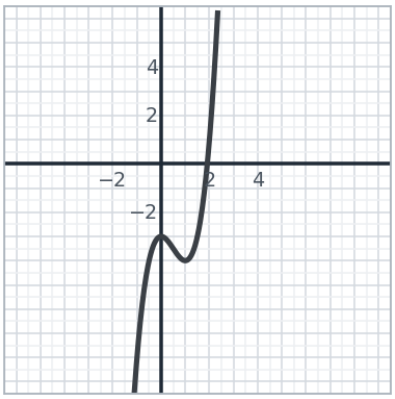
B)

x	y
10	−13
20	−10
35	5
60	30
80	50

C)



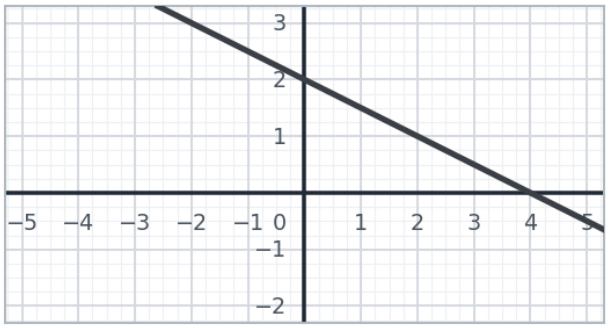
D)



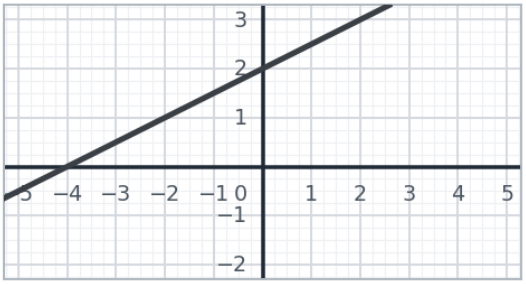
WORKSPACE

Question 7

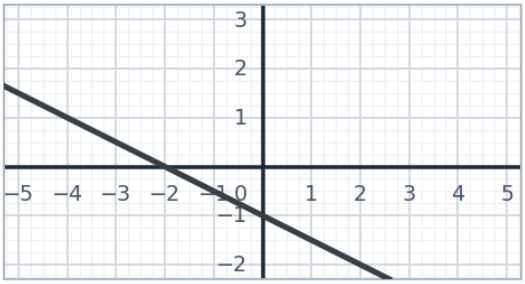
Consider the following graph of the function $y = mx + b$:



Graph A



Graph D



Which of the following options is true in regard to this graph?

A) Graph A is of a function with the same slope m and the same interception point with the y-axis.

B) Function B, $y = \frac{1}{2}x + 2$, has a different slope and the same interception point with the y-axis.

C) Function C, $y = -2x + 2$, has the same slope and the same interception point with the y-axis.

D) Graph D is of a function with a different slope m and a different interception point with the y-axis.

WORKSPACE

Question 8

What is the next number in the following series of numbers?

5 6 11 17 ?

A) 19

B) 23

C) 25

D) 28

WORKSPACE

Question 9

What is the next number in the following series of numbers?

15 10 12 8 10 7 9 ?

A) 6

B) 7

C) 8

D) 9

WORKSPACE

Question 10

What is the next number in the following series of numbers?

2 6 18 54 ?

A) 162

B) 92

C) 102

D) 152

WORKSPACE

Question 11

What is the next number in the following series of numbers?

2 5 11 23 47 ?

A) 70

B) 86

C) 94

D) 95

WORKSPACE

Question 12

What is the next number in the following series of numbers?

50 34 26 22 ?

A) 22

B) 21

C) 20

D) 24

WORKSPACE

Question 13

What is the next number in the following series of numbers?

2 6 3 9 6 ?

A) 12

B) 16

C) 18

D) 24

WORKSPACE

Question 14

Simplify the following expression:

$2(3x^2 + 2x^3 - 9 - 4x^2 - 5x + 1 - 3x^3 + x^2)$

A) $-x^3 + 5x^2 + x - 9$

B) $-2x^3 - 10x - 16$

C) $2x^2 - 10x^2 - 20$

D) $x^2 + x - 9$

WORKSPACE

Question 15

Consider the following formula:

$y = 2(2x + 1)(x - 1)$

Which of the following formulas is equivalent to this one?

A) $4x^2 - x - 1$

B) $4x^2 - 2x - 2$

C) $2x^2 - x - 1$

D) $2x^2 - 2x - 2$

WORKSPACE

Question 16

Factor the polynomial:

$x^{-2} + y^3x^3$

A) $x^{-2}(1 + y^3x^5)$

B) $x^{-2}(x^{-2} + y^3x)$

C) $x^{-2}(1 + y^3x)$

D) $x^{-2}y^2(y^2 + yx)$

WORKSPACE

Question 17

Factor the polynomial:

$x^2 + 6x - 27$

A) $(x + 3)(x - 9)$

B) $(x + 8)(x - 2)$

C) $(x + 9)(x - 3)$

D) $(x - 8)(x + 2)$

WORKSPACE

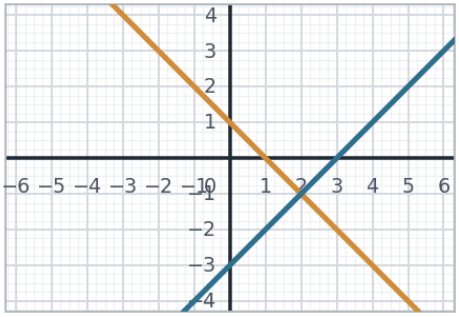
Question 18

Which of the graphs represents the following two linear equations?

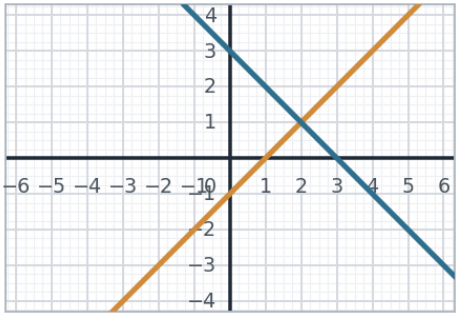
$x + y = 3$

$x - y = 1$

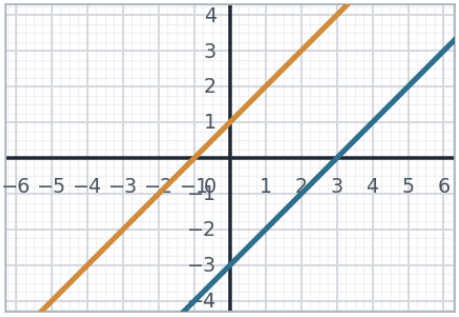
A)



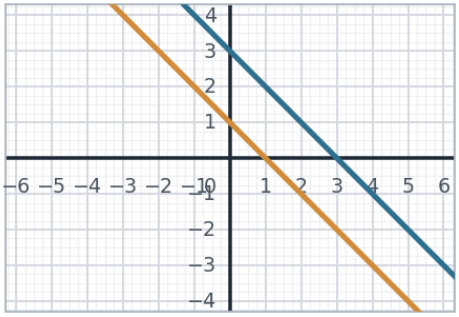
B)



C)



D)



WORKSPACE

Question 19

Consider the following formula:

$A = 7 - B$

Which of the following statements is true for this formula?

A) If A is less than -7, B is positive.

B) If A is greater than 7, B is positive.

C) If B is greater than -7, A is positive.

D) If B is less than -7, A is negative.

WORKSPACE

Question 20

Consider the following formula:

$W = 2P + 2$

Which of the following statements is true for this formula?

A) The difference between the values of W and P changes.

B) The difference between the values of W and P is constant.

C) For positive W values, $P > 0$.

D) For negative P values, $W < 0$.

WORKSPACE

Question 21

Consider the following formula:

$2.5x = 7.5y + 15$

Which of the following statements is true for this formula?

A) When the value of y is less than -2, x is negative.

B) When the value of x is less than 15, y is negative.

C) When the value of x is less than 6, y is positive.

D) When the value of y is greater than -5, x is positive.

WORKSPACE

Question 22

Consider the following formula:

$$\frac{1}{2}x = 2 - \frac{1}{4}y$$

Which of the following statements is true for this formula?

A) When the value of x is less than 4, y is negative.

B) When the value of x is greater than 4, y is positive.

C) When the value of y is less than -6, x is positive.

D) When the value of y is greater than -6, x is positive.

WORKSPACE

Question 23

Consider the following formula:

$x = 0.5y + 6$

Which of the following statements is true for this formula?

A) When the value of y is greater than -6, x is positive.

B) When the value of y is lower than -6, x is negative.

C) When the value of y is greater than -12, x is negative.

D) When the value of y is lower than -12, x is positive.

WORKSPACE

Question 24

Consider the following formula:

$T = G + 1$

Which of the following statements is true for this formula?

A) $G + T > 0$

B) $T + G > 2T$

C) $G - T > 0$

D) $(T - G) > (G - T)$

WORKSPACE

Question 25

Solve the proportion:

$$\frac{k+3}{14-k} = \frac{6}{12}$$

A) $1\frac{1}{3}$

B) $1\frac{2}{3}$

C) 3

D) $2\frac{2}{3}$

WORKSPACE

Question 26

Evaluate the expression:

For $x = 11$ and $y = \frac{3x}{11}$,

$$\frac{4x+1}{4y-3} + \frac{4y-3}{4x+1} = ?$$

A) 4.2

B) 4.4

C) 4.5

D) 5.2

WORKSPACE

Question 27

Evaluate the formula:

$$T = \frac{1}{f}$$

$$f = \frac{w}{2\pi}$$

$T = ?$

A) $\frac{2\pi}{w}$

B) $\frac{w}{2\pi}$

C) $\frac{f}{w}$

D) $\frac{f}{2\pi}$

WORKSPACE

Question 28

Solve the following system of equations:

i. $2x - 4y = 12$

ii. $3x - 2y = 10$

A) $(-2, 2)$

B) $(2, -1)$

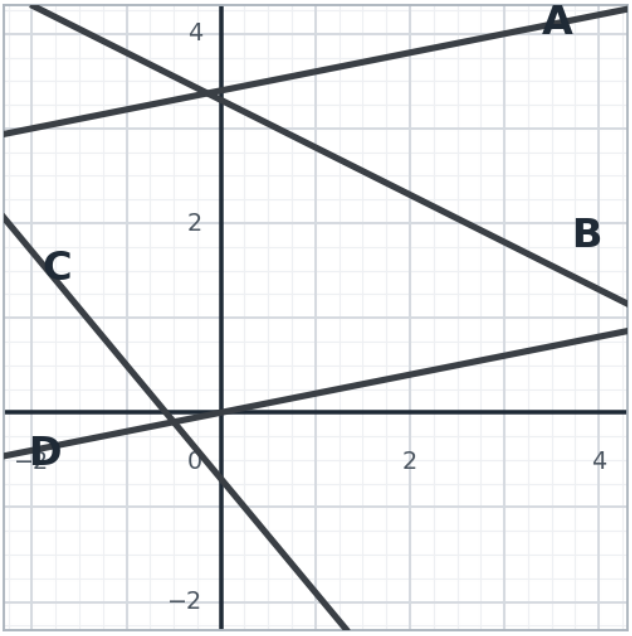
C) $(-2, 1)$

D) $(2, -2)$

WORKSPACE

Question 29

Which of the following graphs represents the function $y = 0.2x$?



A) A

B) B

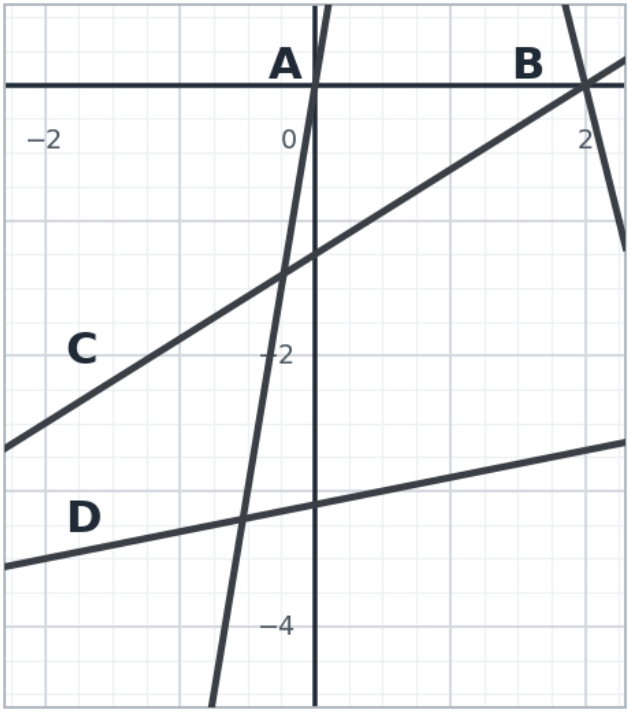
C) C

D) D

WORKSPACE

Question 30

Which of the following graphs represents the function $y = -4x + 8$?



A) A

B) B

C) C

D) D

WORKSPACE

Question 31

Evaluate the value of the unknown variable:

$$\frac{d}{e-f}+\frac{d(f-g)}{e-g}=H$$

$d=18,\ e=9,\ f=3$, and $g=1$. Find H.

A) 7.5

B) 40

C) 7

D) 13.5

WORKSPACE

Question 32

Isolate the variable from the formula:

$$XY = \frac{AB}{C} + \frac{D}{F}$$

$C = ?$

A) $XY - AB + \frac{D}{F}$

B) $XY - AB - \frac{D}{F}$

C) $\frac{XY - \frac{D}{F}}{AB}$

D) $\frac{AB}{XY - \frac{D}{F}}$

WORKSPACE

Question 33

Isolate the variable from the formula:

$$A = \frac{d_1 d_2}{2}$$

$d_1 = ?$

A) $2A - d_2$

B) $\frac{2A}{d_2}$

C) $\frac{d_2}{2A}$

D) $\frac{Ad_2}{2}$

WORKSPACE

Answer Key

1. B	**10. A**	**19. A**	**28. D**
2. D	**11. D**	**20. A**	**29. D**
3. A	**12. C**	**21. A**	**30. B**
4. C	**13. C**	**22. C**	**31. A**
5. A	**14. B**	**23. A**	**32. D**
6. C	**15. B**	**24. D**	**33. B**
7. B	**16. A**	**25. D**	
8. D	**17. C**	**26. D**	
9. B	**18. B**	**27. A**	

1. B $10B - 30 = 50 - B - 2B - 2 \rightarrow 10B - 30 = 48 - 3B \rightarrow 13B = 78 \rightarrow B = 6$

2. D $3A - 16 - 2 = 30 + A \rightarrow 2A = 48 \rightarrow A = 24$

3. A $\frac{A^3 + A}{A} = A^2 + 1$. $(4)(6) = 24$. $A^2 + 1 + 24 - A^2 = 25$

4. C $-1 + 1 = 0$, $-10 + 1 = -9$, $-100 + 1 = -99$ ✓
Table A is $y = -x - 1$. D works out to $y = 1$.

5. A $\frac{10x}{5} = 2x$. Every row of table A is 2 times x ✓

6. C Every y is 3 less than x: $y = x - 3$. The graph crosses at (0, −3) and (3, 0) ✓
A means $x - (-3) = x + 3$. Table B: $10 - 3 = 7$, not −13.

7. B Given line: crosses at (0, 2) and (4, 0) $\rightarrow y = -\frac{1}{2}x + 2$.
$y = \frac{1}{2}x + 2$: different slope, same y-intercept ✓
Graph D has the same slope ($-\frac{1}{2}$), so 'different slope' is false.

8. D Each number is the sum of the two before it: $11 + 17 = 28$

9. B Steps: −5, +2, −4, +2, −3, +2 ... next is −2: $9 - 2 = 7$

10. A Times 3 each time: $54 \times 3 = 162$

11. D Double and add 1: $47 \times 2 + 1 = 95$

12. C Differences: −16, −8, −4 (each half the last). Next is −2: $22 - 2 = 20$

Answer Key (cont.)

13. C Alternates ×3 then −3: $6 \times 3 = 18$

14. B Inside first: $-x^3 + 0x^2 - 5x - 8$ (the x^2 terms cancel).
Times 2: $-2x^3 - 10x - 16$

15. B $(2x+1)(x-1) = 2x^2 - x - 1$. Times 2: $4x^2 - 2x - 2$

16. A Pull out x^{-2}: $x^3 \div x^{-2} = x^{3+2} = x^5$. So $x^{-2}(1 + y^3x^5)$

17. C Two numbers that multiply to −27 and add to +6: +9 and −3.

18. B Get y alone: $y = 3 - x$ (falls, crosses at 3) and $y = x - 1$ (rises, crosses at −1).
They meet at (2, 1). That's graph B.

19. A $B = 7 - A$. If $A < -7$, then $B > 14$ — positive ✓
B: $A > 7$ makes B negative. D: $B < -7$ makes $A > 14$.

20. A $W - P = P + 2$, which changes as P changes ✓
C and D both fail at $P = -\frac{1}{2}$ ($W = 1$).

21. A Divide by 2.5: $x = 3y + 6$, which is 0 at $y = -2$. Below −2, x is negative ✓

22. C Times 2: $x = 4 - \frac{1}{2}y$, which is 0 at $y = 8$. x is positive whenever $y < 8$.
So $y < -6 \rightarrow$ positive ✓. D fails at $y = 10$ ($x = -1$).

23. A $x = 0$ at $y = -12$. Above −12, x is positive — so above −6 it's positive too ✓
B fails at $y = -8$ ($x = 2$).

24. D $T - G = 1$ and $G - T = -1$, so $1 > -1$ is always true ✓
B: $T + G = 2T - 1$, which is less than $2T$.

25. D $\frac{6}{12} = \frac{1}{2}$. $2(k+3) = 14 - k \rightarrow 3k = 8 \rightarrow k = 2\frac{2}{3}$

26. D $y = 3$. $\frac{45}{9} + \frac{9}{45} = 5 + 0.2 = 5.2$

27. A $T = 1 \div \frac{w}{2\pi}$ — dividing by a fraction flips it: $\frac{2\pi}{w}$

Answer Key (cont.)

28. D Double ii: $6x - 4y = 20$. Subtract i: $4x = 8 \rightarrow x = 2$. Then $4 - 4y = 12 \rightarrow y = -2$

29. D $y = 0.2x$ goes through (0, 0) and rises slowly (up 1 for every 5 right).
Only D passes through the origin. A has the same slope but starts high.

30. B $y = 0$ at $x = 2$, and it falls steeply (down 4 per 1).
Only B passes through (2, 0) and falls.

31. A $\frac{18}{6} + \frac{18 \times 2}{8} = 3 + 4.5 = 7.5$

32. D Subtract $\frac{D}{F}$: $\frac{AB}{C} = XY - \frac{D}{F}$. Flip both sides and multiply by AB.

33. B Multiply by 2: $2A = d_1 d_2$. Divide by d_2.